```
                HH        HH   AAAAAAAAAA   NN        NN  OOOOOOOOOOOO   IIIIIIIIII    CCCCCCCCCC   OOOOOOOOOOOO  BBBBBBBBBB
               HH        HH  AAAAAAAAAAAA  NNN       NN  OOOOOOOOOOOO   IIIIIIIIII   CCCCCCCCCCCC  OOOOOOOOOOOO  BBBBBBBBBBB
              HH        HH  AA        AA  NNNN      NN  OO        OO       II       CC        CC  OO        OO  BB        BB
             HH        HH  AA        AA  NN NN     NN  OO        OO       II       CC            OO        OO  BB        BB
            HH        HH  AA        AA  NN  NN    NN  OO        OO       II       CC            OO        OO  BB       BB
           HHHHHHHHHHHH  AAAAAAAAAAAA  NN   NN   NN  OO        OO       II       CC            OO        OO  BBBBBBBBB
          HHHHHHHHHHHH  AAAAAAAAAAAA  NN    NN  NN  OO        OO       II       CC            OO        OO  BBBBBBBBB
         HH        HH  AA        AA  NN     NN NN  OO        OO       II       CC            OO        OO  BB       BB
        HH        HH  AA        AA  NN      NNNN  OO        OO       II       CC            OO        OO  BB        BB
       HH        HH  AA        AA  NN       NNN  OO        OO       II       CC        CC  OO        OO  BB        BB
      HH        HH  AA        AA  NN        NN  OOOOOOOOOOOO   IIIIIIIIII   CCCCCCCCCCCC  OOOOOOOOOOOO  BBBBBBBBBBB
     HH        HH  AA        AA  NN         N  OOOOOOOOOOOO   IIIIIIIIII    CCCCCCCCCC   OOOOOOOOOOOO  BBBBBBBBBB


                  JJJJJJJJJJ   9999999999   5555555555555       11                                    AAAAAAAAAA
                  JJJJJJJJJJ  99999999999  5555555555555      111                                   AAAAAAAAAAAA
                      JJ      99        99  55                1111                                   AA        AA
                      JJ      99        99  55                  11                                   AA        AA
                      JJ      99        99  55                  11                                   AA        AA
                      JJ      99999999999  555555555           11                                   AAAAAAAAAAAA
                      JJ      99999999999  5555555555          11                                   AAAAAAAAAAAA
                      JJ                99           55        11                                   AA        AA
                JJ    JJ                99            55       11                                   AA        AA
                JJ    JJ      99        99            55       11                                   AA        AA
                JJJJJJJ       99999999999  5555555555555   1111111111                              AA        AA
                 JJJJJJ        9999999999   55555555555    1111111111                              AA        AA


****A  START  JOB  951  HANOICOB  Hanoi iter COBOL      ROOM        8.46.41 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  951  START  A****
****A  START  JOB  951  HANOICOB  Hanoi iter COBOL      ROOM        8.46.41 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  951  START  A****
****A  START  JOB  951  HANOICOB  Hanoi iter COBOL      ROOM        8.46.41 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  951  START  A****
```

J E S 2 J O B L O G

```
20.46.40 JOB  951  $HASP373 HANOICOB STARTED - INIT  1 - CLASS A - SYS TK4-
20.46.40 JOB  951  IEF403I HANOICOB - STARTED - TIME=20.46.40
20.46.40 JOB  951  IEFACTRT - Stepname  Procstep  Program   Retcode
20.46.40 JOB  951  HANOICOB   HANOI     COB       IKFCBL00  RC= 0004
20.46.41 JOB  951  HANOICOB   HANOI     GO        LOADER    RC= 0000
20.46.41 JOB  951  IEF404I HANOICOB - ENDED - TIME=20.46.41
20.46.41 JOB  951  $HASP395 HANOICOB ENDED


------ JES2 JOB STATISTICS ------

 01 NOV 26 JOB EXECUTION DATE

       213 CARDS READ

       353 SYSOUT PRINT RECORDS

         0 SYSOUT PUNCH RECORDS

      0.00 MINUTES EXECUTION TIME
```

```
1       //HANOICOB JOB (MBA),                                                  JOB  951
        //             'Hanoi iter COBOL',
        //             CLASS=A,
        //             MSGCLASS=A,
        //             NOTIFY=HERC01,
        //             REGION=8M,TIME=1440,
        //             MSGLEVEL=(1,1),
        //            USER=HERC01,PASSWORD=            GENERATED BY GDL
        ********************************************************************
        ***
        *** Name: HERC01.MYSTUFF.JCL(HANOICOB)
        ***
        *** Desc: Towers of Hanoi.
        ***       Number of slices n (PIC 99, 1<=n<=10), source stack s
        ***       (PIC 99, 0<=s<=2) and destination stack d (PIC 99,
        ***       (0<=d<=2)&(d<>s)) are entered in //GO.SYSIN DD
        ***
        ********************************************************************
2       //HANOI    EXEC COBUCG,
        //         PARM.COB='FLAGW,LOAD,SUPMAP,SIZE=2048K,BUF=1024K'
3       XXCOBUCG PROC SOUT='*'                                                00000100
4       XXCOB EXEC PGM=IKFCBL00,                                              00000200
        XX         PARM='LOAD,SIZE=2048K,BUF=1024K'                           00000300
5       XXSYSPRINT DD SYSOUT=&SOUT                                            00000400
6       XXSYSUT1 DD UNIT=SYSDA,SPACE=(460,(700,100))                          00000500
7       XXSYSUT2 DD UNIT=SYSDA,SPACE=(460,(700,100))                          00000600
8       XXSYSUT3 DD UNIT=SYSDA,SPACE=(460,(700,100))                          00000700
9       XXSYSUT4 DD UNIT=SYSDA,SPACE=(460,(700,100))                          00000800
10      XXSYSLIN DD DSNAME=&LOADSET,DISP=(MOD,PASS),                          00000900
        XX             UNIT=SYSDA,SPACE=(80,(500,100))                        00001000
11      //COB.SYSPUNCH DD DUMMY
12      //COB.SYSIN    DD *
13      //COB.SYSLIB  DD DSNAME=SYS1.COBLIB,DISP=SHR
14      XXGO EXEC PGM=LOADER,PARM='MAP,LET',COND=(5,LT,COB)                   00001100
15      XXSYSLIN DD DSNAME=*.COB.SYSLIN,DISP=(OLD,DELETE)                     00001200
16      XXSYSLOUT DD SYSOUT=&SOUT                                             00001300
17      XXSYSLIB DD DSNAME=SYS1.COBLIB,DISP=SHR                               00001400
18      //GO.SYSOUT   DD SYSOUT=*,DCB=(RECFM=FBA,LRECL=161,BLKSIZE=16100)
19      //GO.SYSIN    DD *
```

```
 STMT NO. MESSAGE
-
    5      IEF653I SUBSTITUTION JCL - SYSOUT=*
   16      IEF653I SUBSTITUTION JCL - SYSOUT=*
IEF236I ALLOC. FOR HANOICOB COB HANOI
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I 170  ALLOCATED TO SYSUT1
IEF237I 140  ALLOCATED TO SYSUT2
IEF237I 190  ALLOCATED TO SYSUT3
IEF237I 180  ALLOCATED TO SYSUT4
IEF237I 180  ALLOCATED TO SYSLIN
IEF237I DMY  ALLOCATED TO SYSPUNCH
IEF237I JES2 ALLOCATED TO SYSIN
IEF237I 148  ALLOCATED TO SYSLIB
IEF142I HANOICOB COB HANOI - STEP WAS EXECUTED - COND CODE 0004
IEF285I   JES2.JOB00951.SO0103                          SYSOUT
IEF285I   SYS26305.T204640.RA000.HANOICOB.R0000001      DELETED       *--------6
IEF285I   VOL SER NOS= WORK01.
IEF285I   SYS26305.T204640.RA000.HANOICOB.R0000002      DELETED       *--------6
IEF285I   VOL SER NOS= WORK00.
IEF285I   SYS26305.T204640.RA000.HANOICOB.R0000003      DELETED       *--------9
IEF285I   VOL SER NOS= WORK03.
IEF285I   SYS26305.T204640.RA000.HANOICOB.R0000004      DELETED       *--------5
IEF285I   VOL SER NOS= WORK02.
IEF285I   SYS26305.T204640.RA000.HANOICOB.LOADSET       PASSED        *-------79
IEF285I   VOL SER NOS= WORK02.
IEF285I   JES2.JOB00951.SI0101                          SYSIN
IEF285I   SYS1.COBLIB                                   KEPT          *--------0
IEF285I   VOL SER NOS= MVSRES.
IEF373I STEP /COB      / START 26305.2046
IEF374I STEP /COB      / STOP  26305.2046 CPU    0MIN 00.14SEC SRB    0MIN 00.06SEC VIRT  2076K SYS   216K
*********************************************************************************************************************
*     1. Jobstep of job: HANOICOB    Stepname: COB          Program name: IKFCBL00   Executed on 01.11.26 from 20.46.40 to 20.46.40 *
*         elapsed time  00:00:00,30                    CPU-Identifier:  TK4-           Page-in:      0                          *
*             CPU time  00:00:00,20              Virtual Storage used:   2076K         Page-out:      0                          *
*           corr. CPU:  00:00:00,20   CPU time has been corrected by  1 / 1,0  multiplier                                      *
*                                                                                                                               *
*     I/O Operation                                                                                                             *
*     Number of records read via DD * or DD DATA:    185                                                                        *
*     DMY.......0 170.......6 140.......6 190.......9 180.......5 180......79 DMY.......0 DMY.......0 148.......0              *
*                                                                                                                               *
*                                      Charge for step (w/o SYSOUT):          0,33                                          *
*********************************************************************************************************************
IEF236I ALLOC. FOR HANOICOB GO HANOI
IEF237I 180  ALLOCATED TO SYSLIN
IEF237I JES2 ALLOCATED TO SYSLOUT
IEF237I 148  ALLOCATED TO SYSLIB
IEF237I JES2 ALLOCATED TO SYSOUT
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I HANOICOB GO HANOI - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS26305.T204640.RA000.HANOICOB.LOADSET       DELETED       *-------80
IEF285I   VOL SER NOS= WORK02.
IEF285I   JES2.JOB00951.SO0104                          SYSOUT
IEF285I   SYS1.COBLIB                                   KEPT          *-------47
IEF285I   VOL SER NOS= MVSRES.
IEF285I   JES2.JOB00951.SO0105                          SYSOUT
IEF285I   JES2.JOB00951.SI0102                          SYSIN
IEF373I STEP /GO       / START 26305.2046
IEF374I STEP /GO       / STOP  26305.2046 CPU    0MIN 00.05SEC SRB    0MIN 00.01SEC VIRT   332K SYS   204K
*********************************************************************************************************************
*     2. Jobstep of job: HANOICOB    Stepname: GO           Program name: LOADER     Executed on 01.11.26 from 20.46.40 to 20.46.41 *
*         elapsed time  00:00:00,09                    CPU-Identifier:  TK4-           Page-in:      0                          *
*             CPU time  00:00:00,06              Virtual Storage used:    332K         Page-out:      0                          *
```

```
*           corr. CPU:  00:00:00,06   CPU time has been corrected by  1 / 1,0  multiplier                                  *
*                                                                                                                        *
*     I/O Operation                                                                                                      *
*     Number of records read via DD * or DD DATA:      1                                                                 *
*     180......80 DMY.......0 148......47 DMY.......0 DMY.......0                                                        *
*                                                                                                                        *
*                                          Charge for step (w/o SYSOUT):          0,10                                *
************************************************************************************************************************
IEF375I  JOB /HANOICOB/ START 26305.2046
IEF376I  JOB /HANOICOB/ STOP  26305.2046 CPU    0MIN 00.19SEC SRB    0MIN 00.07SEC
```

```
00001         * ////////////////////////////////////////////////////////
00002         * // Name: ManInBlueAT
00003         * // Program: The Towers of Hanoi
00004         * // Language: COBOL
00005         * //
00006         * // Desc:
00007         * // Move n slices from source stack s to destination stack d
00008         * // using help stack h (3-s-d)
00009         * // As recursion is not supported in COBOL 360 an iterative
00010         * // algorithm is applied. Please refer to
00011         * // https://en.wikipedia.org/wiki/Tower_of_Hanoi
00012         * //
00013         * // Changes:
00014         * // - ManInBlueAT, 2024-05-18 New program
00015         * ////////////////////////////////////////////////////////
00016          IDENTIFICATION DIVISION.
00017          PROGRAM-ID.  'HANOI'.
00018          ENVIRONMENT DIVISION.
00019          CONFIGURATION SECTION.
00020          SOURCE-COMPUTER.  IBM-360.
00021          OBJECT-COMPUTER.  IBM-360.
00022          INPUT-OUTPUT SECTION.
00023          FILE-CONTROL.
00024              SELECT HANOI-SYSIN
00025                  ASSIGN TO UT-S-SYSIN.
00026        *
00027          DATA DIVISION.
00028          FILE SECTION.
00029          FD  HANOI-SYSIN
00030              RECORDING MODE IS F
00031              RECORD CONTAINS 80 CHARACTERS
00032              BLOCK  CONTAINS  1 RECORDS
00033              LABEL RECORDS ARE OMITTED
00034              DATA RECORD IS HANOI-SYSIN-RECORD.
00035              01  HANOI-SYSIN-RECORD.
00036                02 HANOI-SYSIN-SLICES PIC 99.
00037                02 HANOI-SYSIN-SOURCE PIC 99.
00038                02 HANOI-SYSIN-DEST   PIC 99.
00039                02 FILLER             PIC X(74).
00040          WORKING-STORAGE SECTION.
00041              77 N            PIC 99.
00042              77 S            PIC 99.
00043              77 D            PIC 99.
00044              77 HLP-1         PIC 99.
00045              77 HLP-2         PIC 99.
00046              77 I             PIC 99999 COMP.
00047              77 STEPS         PIC 99999 COMP.
00048              77 SM            PIC 99.
00049              77 DM            PIC 99.
00050              01 DISPLAY-ZZ.
00051                05 DISPL-ZZ   PIC Z9 OCCURS 4.
00052              01 DISPL-STEPS  PIC ZZZZ9.
00053              01 STACK.
00054                05 STACKSTRUC OCCURS 3 TIMES.
```

```
00055                  10 DISPLAY-NO PIC 99.
00056                  10 COUNT      PIC 99.
00057                  10 SLICE      PIC 99 OCCURS 11 TIMES.
00058           *
00059            PROCEDURE DIVISION.
00060            MAIN-PART.
00061                OPEN INPUT HANOI-SYSIN.
00062                READ HANOI-SYSIN AT END
00063                  DISPLAY '** EOF on SYSIN. No data found. Exit.'
00064                  STOP RUN.
00065                MOVE HANOI-SYSIN-SLICES TO N.
00066                MOVE HANOI-SYSIN-SOURCE TO S.
00067                MOVE HANOI-SYSIN-DEST   TO D.
00068                CLOSE HANOI-SYSIN.
00069           *
00070           * Internally we always move slices
00071           * from stack 0 to 1 with help stack 2 if n is odd and
00072           * from stack 0 to 2 with help stack 1 if n is even
00073           * As the index origin in COBOL is 1 the array indices of
00074           * stack is 1 to 3
00075           *
00076                MOVE S TO DISPLAY-NO (1).
00077                DIVIDE 2 INTO N GIVING HLP-1 REMAINDER HLP-2.
00078                IF HLP-2 = 1
00079                  MOVE D TO DISPLAY-NO (2)
00080                  COMPUTE DISPLAY-NO (3) = 3 - S - D
00081                  MOVE DISPLAY-NO (3) TO HLP-1
00082                ELSE
00083                  MOVE D TO DISPLAY-NO (3)
00084                  COMPUTE DISPLAY-NO (2) = 3 - S - D
00085                  MOVE DISPLAY-NO (2) TO HLP-1.
00086           *
00087                MOVE N TO DISPL-ZZ (1).
00088                MOVE S TO DISPL-ZZ (2).
00089                MOVE D TO DISPL-ZZ (3).
00090                MOVE DISPLAY-NO (HLP-1) TO DISPL-ZZ (4).
00091                DISPLAY 'Moving ', DISPL-ZZ (1), ' slices from stack ',
00092                  DISPL-ZZ (2), ' to stack ', DISPL-ZZ (3),
00093                  ' using help stack ', DISPL-ZZ (4).
00094                DISPLAY ' '.
00095           *
00096           * Initialize stack structure. For every stack, maintain the count
00097           * of slices and track the slices by their position number on the
00098           * stack. Initiallly all slices are on stack 0
00099           *
00100                MOVE N TO COUNT (1).
00101                MOVE 0 TO COUNT (2).
00102                MOVE 0 TO COUNT (3).
00103           *
00104           * The slices are maintained in SLICE per stack. The indices used
00105           * are 2 to 11 (maximum 10 slices). SLICE 1 is set to a value
00106           * (99) higher than any n (max 10) so that any existing slice
00107           * number will always be smaller. The smallest slice has the top
00108           * position on the stack with the highest index
00109           *
00110                MOVE 99 TO SLICE (1, 1).
00111                MOVE 99 TO SLICE (2, 1).
```

```
00112                MOVE 99 TO SLICE (3, 1).
00113           *
00114                COMPUTE I = N + 1.
00115                PERFORM INIT-SLICES UNTIL I LESS  2.
00116           *
00117                COMPUTE STEPS = 2 ** N - 1.
00118                MOVE 1 TO I.
00119                PERFORM MOVE-SLICE UNTIL I GREATER STEPS.
00120           *
00121                DISPLAY ' '.
00122                DISPLAY 'Done.'.
00123                STOP RUN.
00124           *
00125            INIT-SLICES.
00126                COMPUTE SLICE (1, I) = N - I + 2.
00127                MOVE 0 TO SLICE (2, I).
00128                MOVE 0 TO SLICE (3, I).
00129                COMPUTE I = I - 1.
00130           *
00131            MOVE-SLICE.
00132           *
00133           * Do series of three different moves until done: 0 to 1 or 0 to 2
00134           * or 1 to 2 or vice versa
00135           *
00136                DIVIDE 3 INTO I GIVING HLP-1 REMAINDER HLP-2.
00137                IF HLP-2 = 1
00138           *        move 0 to 1 or vice versa
00139                  MOVE 1 TO S
00140                  MOVE 2 TO D
00141                ELSE IF HLP-2 = 2
00142           *        move 0 to 2 or vice versa
00143                  MOVE 1 TO S
00144                  MOVE 3 TO D
00145                ELSE
00146           *        move 1 to 2 or vice versa
00147                  MOVE 2 TO S
00148                  MOVE 3 TO D.
00149                PERFORM MOVE-SLICE-1.
00150                COMPUTE I = I + 1.
00151           *
00152            MOVE-SLICE-1.
00153                COMPUTE HLP-1 = COUNT (S) + 1.
00154                COMPUTE HLP-2 = COUNT (D) + 1.
00155           *
00156           * Determine move direction (s to d or d to s) depending on what
00157           * is legal. Even if COUNT (D) is 0 the comparison will work as
00158           * SLICE (stack, 0 --> 1) is initialized with a value higher
00159           * than any number of slices
00160           *
00161                IF SLICE (S, HLP-1) LESS SLICE (D, HLP-2)
00162                  MOVE S TO SM
00163                  MOVE D TO DM
00164                ELSE
00165                  MOVE D TO SM
00166                  MOVE S TO DM.
00167           *
00168           * do the move
```

```
00169          *
00170               COMPUTE HLP-1 = COUNT (SM) + 1.
00171               COMPUTE HLP-2 = COUNT (DM) + 1.
00172               ADD 1 TO HLP-2.
00173               MOVE SLICE (SM, HLP-1) TO SLICE (DM, HLP-2).
00174               COMPUTE COUNT (DM) = HLP-2 - 1.
00175               MOVE 0 TO SLICE (SM, HLP-1).
00176               SUBTRACT 1 FROM COUNT (SM).
00177          *
00178               MOVE I TO DISPL-STEPS.
00179               MOVE SLICE (DM, HLP-2) TO DISPL-ZZ (1).
00180               MOVE DISPLAY-NO (SM) TO DISPL-ZZ (2).
00181               MOVE DISPLAY-NO (DM) TO DISPL-ZZ (3).
00182               DISPLAY 'Step ', DISPL-STEPS ': Moving slice ',
00183                 DISPL-ZZ (1), ' from stack ', DISPL-ZZ (2),
00184                 ' to stack ', DISPL-ZZ (3).
00185          *
```

```
*STATISTICS*     SOURCE RECORDS =   185     DATA DIVISION STATEMENTS =    23     PROCEDURE DIVISION STATEMENTS =    75
*OPTIONS IN EFFECT*     SIZE = 2097152  BUF = 1048576  LINECNT = 57  SPACE1, FLAGW,   SEQ,   SOURCE
*OPTIONS IN EFFECT*     NODMAP, NOPMAP, NOCLIST,   SUPMAP, NOXREF,   LOAD, NODECK, APOST, NOTRUNC, NOLIB, NOVERB
*OPTIONS IN EFFECT*       ZWB
```

CARD	ERROR MESSAGE	
117	IKF5011I-W	AN INTERMEDIATE RESULT OR A SENDING FIELD MIGHT HAVE ITS HIGH ORDER DIGIT POSITION TRUNCATED.

VS LOADER

OPTIONS USED - PRINT,MAP,LET,CALL,RES,NOTERM,SIZE=307200,NAME=**GO

NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR
HANOI	SD	AC010	ILBOSTP0*	SD	ACE48	ILBOSTP1*	LR	ACE5E	ILBODSP0*	SD	ACE80	ILBOXPR0*	SD	AD580
ILBOIDB0*	SD	AD7D8	ILBOIDB1*	LR	AD7F4	ILBOBID0*	SD	AD840	ILBOBID2*	LR	AD84C	ILBOBID1*	LR	AD860
ILBOXMU0*	SD	AD8A8	ILBOXDI0*	SD	AD958									

TOTAL LENGTH 1A50
ENTRY ADDRESS AC010

```
Moving  4 slices from stack  0 to stack  1 using help stack  2

Step     1: Moving slice  1 from stack  0 to stack  2
Step     2: Moving slice  2 from stack  0 to stack  1
Step     3: Moving slice  1 from stack  2 to stack  1
Step     4: Moving slice  3 from stack  0 to stack  2
Step     5: Moving slice  1 from stack  1 to stack  0
Step     6: Moving slice  2 from stack  1 to stack  2
Step     7: Moving slice  1 from stack  0 to stack  2
Step     8: Moving slice  4 from stack  0 to stack  1
Step     9: Moving slice  1 from stack  2 to stack  1
Step    10: Moving slice  2 from stack  2 to stack  0
Step    11: Moving slice  1 from stack  1 to stack  0
Step    12: Moving slice  3 from stack  2 to stack  1
Step    13: Moving slice  1 from stack  0 to stack  2
Step    14: Moving slice  2 from stack  0 to stack  1
Step    15: Moving slice  1 from stack  2 to stack  1

Done.
```

```
                HH        HH   AAAAAAAAAA   NN        NN  OOOOOOOOOOOO   IIIIIIIIII    CCCCCCCCCC   OOOOOOOOOOOO  BBBBBBBBBB
               HH        HH  AAAAAAAAAAAA  NNN       NN  OOOOOOOOOOOO   IIIIIIIIII   CCCCCCCCCCCC  OOOOOOOOOOOO  BBBBBBBBBBB
              HH        HH  AA        AA  NNNN      NN  OO        OO       II       CC         CC  OO        OO  BB        BB
             HH        HH  AA        AA  NN NN     NN  OO        OO       II       CC             OO        OO  BB        BB
            HH        HH  AA        AA  NN  NN    NN  OO        OO       II       CC             OO        OO  BB       BB
           HHHHHHHHHHHH  AAAAAAAAAAAA  NN   NN   NN  OO        OO       II       CC             OO        OO  BBBBBBBBB
          HHHHHHHHHHHH  AAAAAAAAAAAA  NN    NN  NN  OO        OO       II       CC             OO        OO  BBBBBBBBB
         HH        HH  AA        AA  NN     NN NN  OO        OO       II       CC             OO        OO  BB       BB
        HH        HH  AA        AA  NN      NNNN  OO        OO       II       CC             OO        OO  BB        BB
       HH        HH  AA        AA  NN       NNN  OO        OO       II       CC         CC  OO        OO  BB        BB
      HH        HH  AA        AA  NN        NN  OOOOOOOOOOOO   IIIIIIIIII   CCCCCCCCCCCC  OOOOOOOOOOOO  BBBBBBBBBBB
     HH        HH  AA        AA  NN         N  OOOOOOOOOOOO   IIIIIIIIII    CCCCCCCCCC   OOOOOOOOOOOO  BBBBBBBBBB


                JJJJJJJJJJ   9999999999   5555555555555       11                                  AAAAAAAAAA
                JJJJJJJJJJ  999999999999  5555555555555      111                                 AAAAAAAAAAAA
                    JJ      99        99  55                1111                                 AA        AA
                    JJ      99        99  55                  11                                 AA        AA
                    JJ      99        99  55                  11                                 AA        AA
                    JJ      999999999999  555555555           11                                 AAAAAAAAAAAA
                    JJ      999999999999  5555555555          11                                 AAAAAAAAAAAA
                    JJ                99           55         11                                 AA        AA
                JJ  JJ                99            55        11                                 AA        AA
                JJ  JJ      99        99            55        11                                 AA        AA
                JJJJJJJ     999999999999  5555555555555  1111111111                             AA        AA
                 JJJJJJ      9999999999   55555555555   1111111111                             AA        AA


****A   END   JOB  951  HANOICOB  Hanoi iter COBOL      ROOM        8.46.41 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  951   END   A****
****A   END   JOB  951  HANOICOB  Hanoi iter COBOL      ROOM        8.46.41 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  951   END   A****
****A   END   JOB  951  HANOICOB  Hanoi iter COBOL      ROOM        8.46.41 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  951   END   A****
```